

SOFTWARE ENGINEERING 381 - SEN381

ASSIGNMENT 1

Research Foundations for Software Engineering Decisions

Assessment type	Team research assignment
Team size	3 students - same team used for the integrated project
Duration	3 days
Total marks	50
Submission	One concise professional engineering research brief per team
Project relationship	Pre-project research; CivicConnect project decisions are NOT made in this assignment
Referencing	Institution-prescribed academic referencing style
Academic year	2026

Central principle

Research first so that later project decisions are made from evidence, constraints and lifecycle awareness - not from familiarity, popularity or unverified AI recommendations.

1. Purpose of the Assignment

Assignment 1 establishes the research and decision-thinking foundation students will use when the CivicConnect project begins. It is intentionally concise and must be completed within three days. The emphasis is therefore on focused research, synthesis, comparison and engineering reasoning rather than long descriptive writing.

- Distinguish Software Engineering decision-making from simply producing functioning code.
- Recognize decisions and concerns that must be considered early even when implementation occurs later in the lifecycle.
- Research how technology-stack choices should be evaluated against project context and constraints.
- Establish responsible use of generative AI as an engineering assistant while preserving human verification and accountability.
- Practice using credible evidence to support engineering reasoning before formal project decisions are made.

Important boundary

Assignment 1 may inform later project thinking, but it does not constitute CivicConnect requirements, architecture, technology-stack selection, deployment selection or any other project decision. Those decisions must be made later against the project's own approved requirements, constraints, risks and stakeholder needs.

1.1 Prior Learning Applied and SEN381 Progression

Assignment 1 builds on knowledge students have already developed in earlier modules. The purpose is not to repeat those modules, but to use prior knowledge as the foundation for higher-level Software Engineering research, comparison and engineering judgement.

- **Systems Analysis & Design:** stakeholders, requirements, scope, modelling, system boundaries and the relationship between business needs and system solutions.
- **Programming, Object-Oriented Programming, Web Programming and Databases:** implementation knowledge, data and interface concerns, validation, dependencies and awareness of the practical consequences of technical choices.
- **Problem-solving and project foundations:** problem decomposition, teamwork, prioritisation, constraints, planning and reasoned evaluation of alternatives.

SEN381 progression

At SEN381 level, students must move beyond knowing how to analyse, design or program. They must research why an engineering approach is appropriate, compare viable alternatives, recognise lifecycle consequences, use credible evidence, identify trade-offs and preserve human accountability. The assignment therefore prepares students to make later project decisions from evidence rather than familiarity or preference.

Acronym use: Acronyms and abbreviations used in this assignment follow the Acronyms and Abbreviations table in the SEN381 CivicConnect Master Project Brief. Important terms should still be written in full on first meaningful use where this improves clarity.

2. Learning Focus

This assignment supports the early SEN381 emphasis on Software Engineering foundations, stakeholder and constraint awareness, engineering trade-offs, lifecycle thinking, technology evaluation and responsible AI. It prepares students for Milestone 1 without duplicating Milestone 1 artefacts.

3. Submission Expectations

- Submit one team document answering all four questions in the order given.
- Use concise professional writing supported by tables, diagrams and decision models where these communicate analysis more effectively than prose.
- Use at least 6 credible sources across the assignment. A stronger submission normally uses an appropriate mixture of academic/standards sources, recognised professional sources and official technology documentation.
- Use current evidence for claims about contemporary technology and generative AI.
- AI tools are not academic sources. Any important AI-assisted claim must be independently verified.
- Include a reference list and the AI Research & Verification Record provided in Section 9.
- Each team member must make a meaningful contribution and must understand the complete submission.
- Quality of evidence and reasoning is valued over page count. Excessive descriptive content does not attract additional marks.

4. Mark Allocation

Question	Focus	Marks
Q1	Software Engineering Beyond Programming	10
Q2	Thinking Ahead: Decisions Across the Software Lifecycle	15
Q3	Evidence-Based Technology Stack Decisions	15
Q4	Responsible AI in Software Engineering	10
	TOTAL	50

5. Question 1 - Software Engineering Beyond Programming [10]

Research and critically explain why successful software development requires more than selecting a programming language and producing functioning code.

Your response must show how the following concerns interact rather than merely define them:

- Stakeholder needs and requirements
- Scope
- Schedule
- Cost and resources
- Quality
- Engineering risk

Required evidence

- Use credible research to support the discussion.
- Include ONE researched real-world example in which an engineering decision, assumption or constraint created a later benefit, limitation, rework, technical debt, cost or failure.
- Express the main lesson from the example using the relationship: Decision -> constraint/trade-off -> downstream consequence.

Assessment focus

Criterion	Marks
Understanding SE beyond coding and interaction of constraints	3
Quality and relevance of research evidence	2
Real-world example and consequence analysis	3
Critical synthesis and clarity	2

6. Question 2 - Thinking Ahead: Decisions Across the Software Lifecycle [15]

Software engineers must often consider a decision before the lifecycle activity in which that decision will be fully implemented or its consequences become visible.

Research SIX significant Software Engineering decisions or concerns that should be considered early because postponing them may constrain later choices, increase cost/risk or cause rework. Do not simply copy a generic list: select concerns supported by your research and explain why they matter.

For each of the six decisions/concerns, show:

- What the decision or concern is?
- When it should first be considered?
- When it is likely to be implemented or become operationally significant?
- What information is needed to make a sound decision?
- Which later activities it enables, constrains or affects?
- What could happen if consideration is postponed until implementation time?
- Which credible evidence supports the analysis?

Required visual: Decision Horizon Map

Create ONE original visual map that positions the six selected concerns across a software lifecycle. The map must distinguish early consideration from later implementation/effect and show meaningful downstream relationships. It must be explained briefly in the text.

Critical question

Why might waiting until an activity is due for implementation already be too late to make a good engineering decision?

Assessment focus

Criterion	Marks
Selection and relevance of six lifecycle decisions/concerns	3
Analysis of timing, dependencies and information needs	4
Consequences of delayed consideration	3
Decision Horizon Map: accuracy, relationships and clarity	3
Research evidence and synthesis	2

7. Question 3 - Evidence-Based Technology Stack Decisions [15]

Research and compare TWO viable contemporary technology-stack approaches that could be used to develop a modern web-based information system. The purpose is to investigate how a technology decision should be made, not to select a technology stack for CivicConnect.

Required comparison criteria

- Team capability and learning curve

- Security ecosystem
- Maintainability
- Testing and tooling support
- Deployment/platform compatibility
- Development and likely operational cost
- Future change and maintainability implications

Required decision evidence

- Briefly identify the main components/characteristics of each stack only to the extent needed for the comparison.
- Create a weighted decision matrix using the required criteria. You may add a justified criterion if it materially improves the comparison.
- Justify the weights. Do not assign numbers without explaining why a criterion deserves its relative importance.
- Support important scores/claims with credible evidence rather than familiarity or popularity.
- Explain limitations of the matrix and why context can change the result.

Critical question

Why is the highest-scoring or most popular technology stack not automatically the correct technology for every software project?

Assessment focus

Criterion	Marks
Appropriate research of two viable alternatives	3
Comparison against required lifecycle criteria	4
Weighted decision matrix and justified weights	4
Evidence quality and support for claims/scores	2
Critical conclusion and limitations	2

8. Question 4 - Responsible AI in Software Engineering [10]

Research how generative AI can assist contemporary Software Engineering while preserving professional accountability. Keep the response focused: this question is about engineering value, risk and verification, not a catalogue of AI tools.

Required response

- Identify and explain THREE valuable uses of generative AI across Software Engineering activities such as requirements, architecture/design, coding, testing, review or documentation.
- Identify and explain THREE significant risks or limitations associated with AI-assisted engineering.
- Propose a concise human verification/control process that should occur before material AI-generated output becomes part of a controlled software product.
- Answer the accountability question below using evidence and engineering reasoning.

Accountability question

If an engineer accepts AI-generated output that later introduces a defect, vulnerability or incorrect engineering decision, who remains accountable and why?

Assessment focus

Criterion	Marks
Three valuable uses with engineering relevance	2
Three risks/limitations with consequences	3
Human verification/control process	3
Accountability reasoning and evidence	2

9. Research, Referencing and AI Evidence Standard

9.1 Research quality

- Use sources that are credible for the claim being made. Official technology documentation is appropriate for technology capabilities; academic/standards/professional evidence is preferable for broader engineering principles and evaluated claims.
- Do not use marketing claims as independent proof of quality, security, scalability or suitability.
- Distinguish researched fact from team interpretation.
- Paraphrase and synthesise rather than constructing the assignment from long quotations.
- Apply the institution-prescribed referencing style consistently in-text and in the final reference list.

9.2 AI Research & Verification Record

If AI is used, include a short appendix using the following structure. The record is not awarded marks simply for existing; poor verification can reduce the quality of the relevant answer.

Team member	AI tool / use	Purpose	Output used?	How independently verified	What was changed/rejected

9.3 Begin the Evidence Habit Now

Assignment 1 is research rather than a project milestone, but the team should already begin the progressive evidence habits required throughout SEN381. Because the same team continues into CivicConnect, students are strongly encouraged to use the team repository from Assignment 1 to preserve authentic evidence of research, document development, contribution and review.

- Commit appropriate working artefacts regularly rather than uploading a completed document only at the end.
- Use meaningful commit messages and retain visible progression of research, analysis, diagrams, decision models and document revision.
- Do not reconstruct contribution history immediately before submission or a later milestone presentation.
- Assignment 1 does not require the full Milestone 1 GitHub governance regime unless instructed by the lecturer; formal protected-main, Pull Request and approval controls apply when the project-controlled baseline begins.
- The purpose is to establish the professional habit that engineering history and evidence are created while work is being done, not after the fact.

10. Team Accountability

The assignment is completed in the same teams of three established for the project. The submission is shared, but all members are responsible for its integrity and must understand the complete work.

- Dividing the four questions among members does not remove collective responsibility for the final submission.
- The team should retain reasonable evidence of contribution, research activity and document revision.
- Lecturers may ask brief verification questions where contribution or understanding is unclear.
- The research assignment should begin the professional team culture expected later in the project: evidence, review, accountability and responsible AI use.

11. What Assignment 1 Must NOT Become

Do not	Instead
Write a long textbook-style collection of definitions.	Analyse relationships, trade-offs and consequences.
Select the CivicConnect technology stack.	Research how technology-stack decisions should be evaluated.
Create CivicConnect requirements, architecture or project	Build research knowledge that can later inform project work.

artefacts.	
Treat AI output as evidence.	Verify AI-assisted claims using credible sources or technical evidence.
Add pages to appear comprehensive.	Use concise prose, tables, diagrams and decision models to communicate reasoning.

12. Submission Checklist

- ☐ One team document submitted.
- ☐ All four questions answered.
- ☐ Q1 includes one researched real-world consequence example.
- ☐ Q2 includes six researched lifecycle decisions/concerns and one original Decision Horizon Map.
- ☐ Q3 compares two technology stacks and includes a justified weighted decision matrix.
- ☐ Q4 includes three AI uses, three risks, a verification process and accountability response.
- ☐ At least six credible sources used across the assignment.
- ☐ In-text citations and reference list are consistent.
- ☐ AI Research & Verification Record included if AI was used.
- ☐ No CivicConnect project decision has been made prematurely in this assignment.

Assignment 1 takeaway

Good Software Engineering does not ask only, “What must we do now?” It also asks, “What must we consider now so that later options remain feasible, safe, affordable and defensible?”

Academic Integrity and Referencing Disclaimer: All submitted work must include appropriate in-text citations and a corresponding reference list. Failure to provide in-text citations and references will result in an automatic mark of zero (0). Plagiarism, fabricated or misrepresented research/evidence, and inappropriate presentation of another person's or AI-generated work as the team's own are subject to the applicable academic-integrity requirements.